PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Impact of cancer diagnoses on the outcomes of patients with COVID-19 – a systematic review and meta-analysis
AUTHORS	Han, Shuting; Zhuang, Qingyuan; Chiang, Jianbang; Tan, Sze Huey; Chua, Gail; Xie, Conghua; Chua, Melvin; Soon, Yu Yang; Yang, Valerie

VERSION 1 – REVIEW

REVIEWER	Lee, Alvin University College London
REVIEW RETURNED	04-Oct-2020

GENERAL COMMENTS	The authors present a systematic review and meta-analysis of the impact of cancer diagnoses on the outcomes of patients with COVID-19. With the ongoing global COVID-19 pandemic and the risks to cancer patients following COVID-19 infection, this article adds to the growing literature in understanding the risks to cancer patients following COVID-19 infection. The authors focused both on death and disease severity as outcomes. Overall, a well written and easy to follow manuscript especially given the heterogeneity of the datasets the authors were working with. Inclusion dates were 1st Nov 2019 to 1st May 2020. Although a substantial amount of work has gone into the manuscript, there is nearly a 4-month gap between cut-off date and the submission date of this manuscript. In the interim, many large cohort studies looking at patients with cancer following COVID-19 infection have been published, including but not limited to: CCC-19, UK Coronavirus Cancer Monitoring Project, TERAVOLT, Oncovid and the Gustave Roussy Cohort. Given the substantial numbers of patients in these studies and ongoing debate regarding effects of chemotherapy on COVID-19 outcomes as seen in the Gustave Roussy cohort and updated results from CCC-19 presented at ESMO 2020, the authors may want to consider updating their paper so as to reflect timely data, especially as in Europe the first peak is over, the systematic review can focus on “first peak” studies which have been published in the last few months. There has also been a recent published systematic review looking specifically at mortality in patients with cancer and COVID-19 (Saini et al., EJC 2020) which has more up-to-date references. Major comments 1. Several studies are referred to as in press, but have now been published. Could preprints please be included in reference list especially as we need to know which studies were included in your systematic review. (see table 1, and reference to TERAVOLT results in discussion)

	2. The discussion refers to several larger studies published after the cut-off date of 1 May 2020 including the UK Coronavirus Cancer Monitoring Project Lancet May 2020 paper. Given that in the discussion it is implied that these larger studies may change the outcome of looking into contribution of systemic anti-cancer treatment, I wonder if the inclusion date should be moved to take into account these large studies. Minor comments 1. Background – Reference 1 was published in Cancer Discovery, not SSRN Electron Journal. 2. Search methods and study selection – 1st line of paragraph – "published clinical trials" – not all were clinical trials. 3. Might be useful to clarify to the reader that haematological malignancies were included in your analysis. 4. Results paragraph 2 – please clarify for the reader what recent systemic therapy is? (chemo, endocrine, how many days before COVID-19 infection, etc?)

REVIEWER	Wu, Gang Huazhong University of Science and Technology
REVIEW RETURNED	28-Oct-2020

GENERAL COMMENTS	1. The small numbers and incomplete death outcomes at the time of reporting restricted the analysis. The authors searched published studies in electronic databases. However, it is possible that some COVID-19 studies are available in preprint only, e.g. in Biorxiv. 2. The limitations of this meta-analysis include the heterogeneity and retrospective nature of the case studies, including reporting and selection biases. A few studies were reported from the same Hospitals and hence there may be overlapping patients although efforts were taken to reduce this. Wish the authors to assess the potential publication bias.

REVIEWER	Kasi, Anup University of Kansas Medical Center
REVIEW RETURNED	14-Dec-2020

GENERAL COMMENTS	I appreciate the authors aim todetermine the proportion of cancer patients with COVID-19, their severity and mortality outcomes through a systematic review and meta-analysis. Some of the aims have been previously studied and published by other authors in other journals. ABSTRACT: The BACKGROUND subheading is missing in the abstract section. METHODS: Are the authors able to search studies in other databases such as EMBASE, ASCO 2020 Virtual Annual Conference, AACR 2020 COVID-19 and Cancer, ESMO conferences. For examples was https://clincancerres.aacrjournals.org/content/26/18_Supplement/PO-082 from AACR meeting. DISCUSSION The manuscript analyzed 230 patients from five case series were identified with treatment history and clinical outcomes. However a larger study has been conducted with more patients with anticancer treatment.

	https://pubmed.ncbi.nlm.nih.gov/33131376/ showed that Active chemotherapy appears to be associated with higher risk of death in cancer patients with COVID-19. This needs to be included in the discussion section. https://pubmed.ncbi.nlm.nih.gov/32864589/ showed that Male sex was associated with a higher risk of severe illness and death attributable to COVID-19. This needs to be included in the discussion section.

REVIEWER	McLoone, Philip University of Glasgow, Public Health
REVIEW RETURNED	28-Dec-2020

GENERAL COMMENTS	This is generally a well written and well presented paper. The main difficulty I had was that the authors use a variable number of studies for each of their analyses. It was difficult to keep track of the number of studies used in each of the pooled analyses. It would be helpful if the authors sign post the number of studies used more clearly. The authors have not clarified whether increased mortality/severity among patients with cancer is due to their older age and the higher proportion of men. The authors should be more clear on the definition of severity - I found it rather vague - admission to ICU can be considered a measure of severity but above a certain threshold of severity or very old age admission may be limited. I would be grateful if the authors define severity definition more clearly. The following details parts of the text and analysis that need modification or further explanation Abstract line 42 “We aim to determine the proportion of cancer patients with COVID19”. This should be the proportion of COVID19 patients with cancer. Pg 4 ln 53 "The results were presented as proportion …" Proportion should be plural. Pg 5 ln 55 "Heterogeneity among" should be "heterogeneity between" studies Pg 5 Patient and Public involvement should be elaborated with ethical statement Pg 5 line 32 senetence starting “Twenty two studies…” does not read well Pg 10 line 34 “However cancer proportion rate” – sentence does not read well – it is a proportion not a rate Pg 10 line 43 “RR 1.41 …” the RR and CI are not the same as those presented earlier for severity of outcome. Table 2 shows 15 studies with mortality for cancer patients Table 3 only documents 13 studies with mortality for cancer patients Supp Figure 2b only shows mortality among patients with cancer in 12 studies It is not clear why the number of studies varies between table 2,3 and supp Fig 2b

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. Alvin Lee, University College London
Comments to the Author:
The authors present a systematic review and meta-analysis of the impact of cancer diagnoses on the outcomes of patients with COVID-19. With the ongoing global COVID-19 pandemic and the risks to cancer patients following COVID-19 infection, this article adds to the growing literature in understanding the risks to cancer patients following COVID-19 infection. The authors focused both on death and disease severity as outcomes. Overall, a well written and easy to follow manuscript especially given the heterogeneity of the datasets the authors were working with.
Inclusion dates were 1st Nov 2019 to 1st May 2020. Although a substantial amount of work has gone into the manuscript, there is nearly a 4-month gap between cut-off date and the submission date of this manuscript. In the interim, many large cohort studies looking at patients with cancer following COVID-19 infection have been published, including but not limited to:
CCC-19, UK Coronavirus Cancer Monitoring Project, TERAVOLT, Oncovid and the Gustave Roussy Cohort.
Given the substantial numbers of patients in these studies and ongoing debate regarding effects of chemotherapy on COVID-19 outcomes as seen in the Gustave Roussy cohort and updated results from CCC-19 presented at ESMO 2020, the authors may want to consider updating their paper so as to reflect timely data, especially as in Europe the first peak is over, the systematic review can focus on "first peak" studies which have been published in the last few months.
There has also been a recent published systematic review looking specifically at mortality in patients with cancer and COVID-19 (Saini et al., EJC 2020) which has more up-to-date references.

Major comments
1. Several studies are referred to as in press, but have now been published. Could preprints please be included in reference list especially as we need to know which studies were included in your systematic review. (see table 1, and reference to TERAVOLT results in discussion)
2. The discussion refers to several larger studies published after the cut-off date of 1 May 2020 including the UK Coronavirus Cancer Monitoring Project Lancet May 2020 paper. Given that in the discussion it is implied that these larger studies may change the outcome of looking into contribution of systemic anti-cancer treatment, I wonder if the inclusion date should be moved to take into account these large studies.
Minor comments
1. Background – Reference 1 was published in Cancer Discovery, not SSRN Electron Journal.
2. Search methods and study selection – 1st line of paragraph – "published clinical trials" – not all were clinical trials.
3. Might be useful to clarify to the reader that haematological malignancies were included in your analysis.
4. Results paragraph 2 – please clarify for the reader what recent systemic therapy is? (chemo, endocrine, how many days before COVID-19 infection, etc?)

Reply: Dear Reviewer 1, we have now included more cancer COVID-19 cohorts and general COVID-19 cohorts in our MA, with expanded literature review that extends to December 2020. We have added CCC19, UKCCMP and others into our analysis.

We have also made changes to our write up taking into account the minor comments. Thank you for the very useful feedback.

Reviewer: 2

Dr. Gang Wu, Huazhong University of Science and Technology
Comments to the Author:
1. The small numbers and incomplete death outcomes at the time of reporting restricted the analysis. The authors searched published studies in electronic databases. However, it is possible that some COVID-19 studies are available in preprint only, e.g. in Biorxiv.

Reply: Dear Reviewer 2, thank you for your comments. We have included selected preprints into our meta-analysis. We do note that there are multiple large studies that are now available among peer reviewed journals and have expanded our included articles to 57.

2. The limitations of this meta-analysis include the heterogeneity and retrospective nature of the case studies, including reporting and selection biases. A few studies were reported from the same Hospitals and hence there may be overlapping patients although efforts were taken to reduce this. Wish the authors to assess the potential publication bias.

Reply: Thank you for your comments. We had previously included the specific publications with its date of reporting and hospital of reporting to ensure that there is no obvious overlap in the cases reported. However we acknowledged this as a limitation in our discussions and also the issue with publication bias.

Reviewer: 3
Dr. Anup Kasi, University of Kansas Medical Center
Comments to the Author:
I appreciate the authors aim to determine the proportion of cancer patients with COVID-19, their severity and mortality outcomes through a systematic review and meta-analysis. Some of the aims have been previously studied and published by other authors in other journals.

ABSTRACT:
The BACKGROUND subheading is missing in the abstract section.

METHODS:
Are the authors able to search studies in other databases such as EMBASE, ASCO 2020 Virtual Annual Conference, AACR 2020 COVID-19 and Cancer, ESMO conferences. For examples was https://clincancerres.aacrjournals.org/content/26/18_Supplement/PO-082 from AACR meeting.

Reply:
Dear Reviewer 3, thank you for your comments. We have included our search studies to most databases but as we require full publication for our inclusion criteria, we did not include conference papers/abstracts but have included some in the discussions.

DISCUSSION
The manuscript analyzed 230 patients from five case series were identified with treatment history and clinical outcomes. However a larger study has been conducted with more patients with anticancer treatment.
https://pubmed.ncbi.nlm.nih.gov/33131376/ showed that Active chemotherapy appears to be associated with higher risk of death in cancer patients with COVID-19. This needs to be included in the discussion section.

https://pubmed.ncbi.nlm.nih.gov/32864589/ showed that Male sex was associated with a higher risk of severe illness and death attributable to COVID-19. This needs to be included in the discussion section.

Reply: Thank you for your comments. We have made changes to our heading. We have also included recent papers (up to 31 Dec 2020) in our MA but unfortunately we did not include conference abstracts/presentations as they have limited clinical information and accessibility and our decision was to have full text papers as an inclusion criteria. Thank you for the above suggestions and we have referenced the 2 papers in the discussion section.

Reviewer: 4
Mr. Philip McLoone, University of Glasgow
Comments to the Author:
This is generally a well written and well presented paper. The main difficulty I had was that the authors use a variable number of studies for each of their analyses. It was difficult to keep track of the number of studies used in each of the pooled analyses. It would be helpful if the authors sign post the number of studies used more clearly.

Reply:
Dear reviewer 4, thank you for the useful comment - we have now included the number of articles that are included in each section of the analysis. The varying numbers unfortunately could not be avoided due to the clinical qn and analysis required. We have attempted to make that clearer.

The authors have not clarified whether increased mortality/severity among patients with cancer is due to their older age and the higher proportion of men.

Reply:
We could not usefully identify this using our IPD and pooled analysis. Our severity analysis included a limited number of papers and hence it was challenging to do a stratified analysis.

The authors should be more clear on the definition of severity - I found it rather vague - admission to ICU can be considered a measure of severity but above a certain threshold of severity or very old age admission may be limited. I would be grateful if the authors define severity definition more clearly.

Reply:
There is substantial heterogeneity in terms of severity definition among the papers that we have included for the MA. It was challenging to have a very strict definition as that would limit the analysis to a very small number. It is currently defined as ICU, mechanical ventilation or respiratory failure requiring oxygenation support. We also included the Chinese, IDSA and WHO definitions as they were similar in the aspect of respiratory failure requiring O2 supplementation

The following details parts of the text and analysis that need modification or further explanation

Abstract line 42 “We aim to determine the proportion of cancer patients with COVID19”. This should be the proportion of COVID19 patients with cancer.
Pg 4 ln 53 "The results were presented as proportion …" Proportion should be plural.
Pg 5 ln 55 "Heterogeneity among" should be "heterogeneity between" studies
Pg 5 Patient and Public involvement should be elaborated with ethical statement
Pg 5 line 32 senetence starting “Twenty two studies…” does not read well
Pg 10 line 34 “However cancer proportion rate” – sentence does not read well – it is a proportion not a rate
Pg 10 line 43 “RR 1.41 …” the RR and CI are not the same as those presented earlier for severity of outcome.
Table 2 shows 15 studies with mortality for cancer patients
Table 3 only documents 13 studies with mortality for cancer patients
Supp Figure 2b only shows mortality among patients with cancer in 12 studies

It is not clear why the number of studies varies between table 2,3 and supp Fig 2b

Reply:
Thank you very much for the specific comments and we have made the amendments to the above points. the differing mortality studies used for the pooled analysis was due to our definition of having at least 10 cancer patients with mortality outcomes to be included in the mortality analysis to avoid biases due to small sample size.
We will amend supplementary figure 2b.

VERSION 2 – REVIEW

REVIEWER	McLoone, Philip University of Glasgow, Public Health
REVIEW RETURNED	09-Jul-2021

GENERAL COMMENTS	This is a complex (due to the limitations of data) but well written paper. The authors have taken on board my previous comments as much as they are able to. The authors clearly discuss the limitations of the work. My only quibble is that on Page 11 line 51 the authors describe the I2 as 'moderate heterogeneity'. I would be tempted to describe this as 'substantial heterogeneity'

VERSION 2 – AUTHOR RESPONSE

Thank you for your kind consideration. We have made the changes to page 11 line 51 from moderate heterogeneity to substantial heterogeneity. We have also added the filers for our pubmed/medline search.

On top of that, we have made changes to table 2 as per editor's comments and cited Supplementary Figure 4a-c.

Thank you and we look forward to hearing back favourably.